

Flat morphisms

SGA 1, Exposé IV

English draft

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About this draft

This is an unofficial English translation of Exposé IV, “Morphismes plats”, from SGA 1. The source volume is A. Grothendieck and M. Raynaud, *Revêtements étales et groupe fondamental* (SGA 1), Séminaire de géométrie algébrique du Bois Marie, 1960–61. It follows the slightly corrected SMF recomposition, arXiv:math/0206203v2.

This draft contains the whole exposé, including proofs, footnotes, diagrams, the updating remarks added in 2003 by M. Raynaud (between bold brackets and marked (MR)) where the exposé has them, and its bibliography. Statement numbering and mathematical notation follow the source. References to other exposés print the source’s numbers. Apparent misprints in the source have been retained; they are recorded in the accompanying README, separately from the translated text. Scholarly proofreading remains outstanding.

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EXPOSÉ IV

Flat morphisms

We give here mainly the flatness properties which have served us in the preceding exposés. A more detailed study will be found in Chapter IV of the “Éléments de Géométrie Algébrique” in preparation¹, where one studies systematically the following situation: X being locally of finite type over Y locally noetherian, and $\mathcal{F}$ coherent on X and Y -flat, to give relations between the properties of Y , those of $\mathcal{F}$, and those of the coherent sheaves induced by $\mathcal{F}$ on the fibers of $X \rightarrow Y$ (from the point of view notably of the dimension, of the cohomological dimension, of the depth etc.). One has in particular a systematic way of obtaining theorems of *Seidenberg* or *Bertini* type (for hyperplane sections). The essential result for the application of flatness methods in this context is the following (which will be proved below): If Y is integral, X of finite type over Y , $\mathcal{F}$ coherent on X , there exists a nonempty open set U of Y such that $\mathcal{F}$ is Y -flat at the points of X above U . A second way, doubtless even more important, in which flatness enters Algebraic Geometry, is *descent theory*: see for example the two exposés of Grothendieck on the subject at the Séminaire Bourbaki². Flatness thus seems to be one of the central technical notions in Algebraic Geometry.

Let us recall that the notion of flatness and faithful flatness was introduced by Serre in GAGA. An account of the following nos. 1 and 2 will also be found in Bourbaki’s Alg. Comm. (which of course, as the title of the book indicates, does not confine itself to the case of commutative base rings)³.

Contrary to the preceding exposés, we do not assume that the rings considered are necessarily noetherian.

1. Sorites on flat modules

A module M over the ring A is called *flat* (or A -flat if one wants to specify A) if the functor

$$T_M: N \longmapsto M \otimes_A N$$

(which is in any case right exact) is *exact*, i.e. transforms monomorphisms into monomorphisms. It amounts to the same to say that the first right derived functor, or all the right derived functors, are zero, i.e. that one has

$$\mathrm{Tor}_1^A(M, N) = 0 \quad \text{for all } N,$$

resp. that one has

$$\mathrm{Tor}_i^A(M, N) = 0 \quad \text{for } i > 0, \text{ all } N.$$

Since the Tor_i commute with inductive limits, it suffices moreover to verify these conditions for N of finite type, and even (taking then a composition series of N with monogenous quotients) that one has

$$\mathrm{Tor}_1^A(M, N) = 0$$

if N is monogenous, i.e. of the form A/I , I being an ideal of A . Let us note moreover that

$$\mathrm{Tor}_1^A(M, A/I) = 0 \quad \Longleftrightarrow \quad I \otimes_A M \rightarrow M = A \otimes_A M \text{ is } \textit{injective},$$

¹Cf. EGA IV 11 and 12.

²and, for a more detailed account, Exposés VIII and IX below.

³N. Bourbaki, Algèbre Commutative, Chap. I (Modules Plats), Act. Sc. Ind 1290, Paris, Hermann (1961).

as one sees from the exact sequence of the Tor, taking into account $\mathrm{Tor}_1^A(M, A) = 0$. Hence M flat is equivalent to saying that for every ideal I , the natural homomorphism

$$I \otimes_A M \longrightarrow IM$$

is an isomorphism. It suffices moreover to verify this for I of finite type; a fortiori it suffices to verify that the functor $M \otimes$ is exact on modules of finite type.

As whenever one has an exact functor T , if one identifies, for a subobject N' of N , $T(N')$ with a subobject of $T(N)$, one has

$$T(N' \cap N'') = T(N') \cap T(N'')$$

$$T(N' + N'') = T(N') + T(N'')$$

for two subobjects N', N'' of N .

A direct sum of flat modules, a direct factor of a flat module, is flat. In particular, A being flat, a *free* module, hence also a *projective* module, is flat. The tensor product of two flat modules is flat, and if M is flat over A , then $M \otimes_A B$ is flat over B for every change of base $A \rightarrow B$ (because of the associativity of the tensor product and of the fact that a composite of exact functors is exact). If M is flat over B , B flat over A , then M is flat over A (same reason).

The exact sequence of the Tor, plus the “commutativity” of the Tor, gives:

PROPOSITION 1.1. *Let $0 \rightarrow M' \rightarrow M \rightarrow M'' \rightarrow 0$ be an exact sequence of A -modules, M'' being flat. Then*

- (i) *this sequence remains exact upon tensoring with any A -module N*
- (ii) *for M to be flat, it is necessary and sufficient that M' be so.*

One can therefore say that from the point of view of behavior under tensor products, flat modules are “as good” as free or projective modules (and the exact sequence of 1.1 in particular is “as good” as if it split).

Let S be a multiplicatively stable subset of A ; then $S^{-1}A$ is flat over A , since $S^{-1}A \otimes N = S^{-1}N$ is an exact functor in N . If M is A -flat, then $S^{-1}M = S^{-1}A \otimes M$ is $S^{-1}A$ -flat, the converse being true if $M \rightarrow S^{-1}M$ is an isomorphism, i.e. if the $s \in S$ are bijective in M (because of the transitivity of flatness, $S^{-1}A$ being flat over A). More generally, the case of a morphism of preschemes $X \rightarrow Y$ and of a quasi-coherent sheaf $\mathcal{F}$ on X whose flatness with respect to Y one wants to study leads to the situation with two rings:

PROPOSITION 1.2. *Let $A \rightarrow B$ be a homomorphism of rings, M a B -module, T a multiplicatively stable subset of B .*

- (i) *If M is A -flat, then $T^{-1}M$ is A -flat (hence also $S^{-1}A$ -flat for every multiplicatively stable subset S of A mapping into T).*
- (ii) *Conversely, if $M_{\mathfrak{n}}$ is flat over $A_{\mathfrak{n}}$ for every maximal ideal $\mathfrak{n}$ of B , $M_{\mathfrak{n}}$ is flat over A (or, what amounts to the same, over $A_{\mathfrak{m}}$, where $\mathfrak{m}$ is the prime ideal of A inverse image of $\mathfrak{n}$) then M is A -flat.*

Indeed one has the formula, functorial with respect to the A -module N :

$$T^{-1}M \otimes_A N = T^{-1}(M \otimes_A N)$$

since the two sides are functorially isomorphic to $T^{-1}B \otimes_B M \otimes_B N_{(B)}$ (with $N_{(B)} = N \otimes_A B$) by virtue of the associativity formulas of $\otimes$. It follows at once that if $M \otimes_A N$ is exact in N , the same is true of $T^{-1}M \otimes_A N$ (as a composite of two exact functors), whence (i). And (ii) follows likewise, since to verify the exactness of a sequence of B -modules, it suffices to verify the exactness of the localizations at all the maximal ideals of B .

PROPOSITION 1.3. (i) *Let M be a flat A -module. If $x \in A$ is not a divisor of 0 in A , it is not a divisor of 0 in M . In particular, if A is integral, M is torsion-free.*

- (ii) Suppose that A is integral and that for every maximal ideal $\mathfrak{m}$ of A , $A_{\mathfrak{m}}$ is principal (for example A a Dedekind ring, or even principal). For the A -module M to be flat, it is necessary and sufficient that it be torsion-free.

One obtains (i) by noting that the homothety x in M is obtained by tensoring with M the homothety x in A . For (ii), one may already assume A principal thanks to 1.2 (ii); one must show that if M is torsion-free, then for every ideal I of A , the injection $I \rightarrow A$ tensored with M is an injection, which means that the generator x of I is not a divisor of 0 in M , O.K.

2. Faithfully flat modules

A functor F from one category into another is called *faithful* if for all X, Y , the map $\text{Hom}(X, Y) \rightarrow \text{Hom}(F(X), F(Y))$ is injective. If it is an additive functor of additive categories, it amounts to the same to say that $F(u) = 0$ implies $u = 0$, and this implies that $F(X) = 0$ implies $X = 0$. For F to be *faithful and exact*, it is necessary and sufficient that the following condition be satisfied: for every sequence $M' \rightarrow M \rightarrow M''$ of morphisms in $\mathcal{C}$, the transformed sequence $F(M') \rightarrow F(M) \rightarrow F(M'')$ is exact *if and only if* the preceding one is. Or again: F is exact, and $F(X) = 0$ implies $X = 0$. (N.B. to be able to speak of exactness, one must assume the categories in question *abelian*). Suppose one has a family (M_i) of nonzero objects of $\mathcal{C}$ such that every nonzero object of $\mathcal{C}$ has a subobject admitting a quotient isomorphic to one of the M_i . Then F faithful and exact is equivalent to: F exact, and $F(M_i) \neq 0$ for all i . If $\mathcal{C}$ is the category of modules over a ring A , one can take for example for (M_i) the family of the $A/\mathfrak{m}$, $\mathfrak{m}$ running through the maximal ideals of A . (Indeed, every nonzero module admits a nonzero monogenous submodule, hence isomorphic to some A/I , I an ideal $\neq A$, which by Krull admits a quotient $A/\mathfrak{m}$). From these sorites, one deduces in particular:

PROPOSITION 2.1. *Let M be an A -module. Equivalent conditions:*

- (i) *The functor $M \otimes_A$ is faithful and exact.*
- (i bis) *M is flat, and $M \otimes_A N = 0$ implies $N = 0$*
- (i ter) *M is flat, and $M \otimes A/\mathfrak{m} \neq 0$ for every maximal ideal $\mathfrak{m}$ of A .*
- (ii) *For every sequence of homomorphisms $N' \rightarrow N \rightarrow N''$, the sequence tensored with M is exact if and only if the initial sequence is.*

One then says that M is a *faithfully flat* A -module. In particular, if M is faithfully flat, then $N \rightarrow N'$ is a monomorphism (epimorphism, isomorphism) if and only if the homomorphism tensored with M is. A faithfully flat module is *faithful*, since the homothety f in M is obtained by tensoring with M the homothety f in A .

One sees as in 1 the usual transitivity properties: the tensor product of two faithfully flat modules is faithfully flat; if M is faithfully flat over A , $M \otimes_A B$ is faithfully flat over B for every extension of the base $A \rightarrow B$; if B is an A -algebra which is faithfully flat over A and if M is a faithfully flat B -module, it is a faithfully flat A -module.

COROLLARY 2.2. *Let $A \rightarrow B$ be a local homomorphism of local rings, M a B -module of finite type. For M to be faithfully flat over A , it is necessary and sufficient that it be flat over A and nonzero.*

This follows from criterion (i ter) and from Nakayama. In particular, for B to be A -flat, it is necessary and sufficient that it be faithfully A -flat.

PROPOSITION 2.3. *Let $A \rightarrow B$ be a homomorphism of rings, M a B -module which is faithfully flat over A . For every prime ideal $\mathfrak{p}$ of A , there exists a prime ideal $\mathfrak{q}$ of B which induces it.*

Dividing by $\mathfrak{p}$, one is reduced to the case where $\mathfrak{p} = 0$. Localizing at the prime ideal 0, one is reduced to the case where A is a field. But M , being faithfully flat over A , is nonzero, a fortiori $B \neq 0$, hence B has a prime ideal, which can only induce the unique prime ideal of A ! Geometrically, one can say that the existence of a quasi-coherent sheaf $\mathcal{F}$ on $X = \text{Spec}(B)$ which is “faithfully flat” relative to A implies that $X \rightarrow Y = \text{Spec}(A)$ is *surjective*.

COROLLARY 2.4. *Suppose M flat over A , of finite type over B and $\text{Supp } M = \text{Spec}(B)$ (i.e. $M_{\mathfrak{q}} \neq 0$ for every prime ideal $\mathfrak{q}$ of B). Then the prime ideals $\mathfrak{q}$ of B minimal among those containing $\mathfrak{p}B$ induce $\mathfrak{p}$.*

One is again reduced to the case $\mathfrak{p} = 0$ (since the hypotheses are all preserved on dividing), hence A integral. One is reduced to the following statement:

COROLLARY 2.5. *M being as above, every minimal prime ideal $\mathfrak{q}$ of B induces a prime ideal $\mathfrak{p}$ of A which is minimal.*

Indeed, localizing at $\mathfrak{p}$ and $\mathfrak{q}$, one is reduced to proving that if A and B are local and the homomorphism $A \rightarrow B$ local, M a nonzero B -module flat over A , and if B is of dimension 0, then A is of dimension 0. By 2.2 and 2.3, one concludes that every prime ideal of A is induced by a prime ideal of B , hence by the maximal ideal of B , hence is the maximal ideal, qed. Geometrically, 2.5 means that every irreducible component of $X = \text{Spec}(B)$ dominates some irreducible component of $Y = \text{Spec}(A)$ (granted the existence of a quasi-coherent sheaf of finite type on X , with support X , and flat with respect to Y).

It will be noted that one did not have to assume in 2.4 M faithfully flat over A , but nothing then guarantees the existence of a prime ideal containing $\mathfrak{p}B$ (hence of a minimal one among such).

PROPOSITION 2.6. *Let $i: A \rightarrow B$ be a homomorphism of rings. Equivalent conditions:*

- (i) *B is a faithfully flat A -module.*
- (ii) *B is flat over A , and $\text{Spec}(B) \rightarrow \text{Spec}(A)$ is surjective*
- (ii bis) *B is flat over A , and every maximal ideal is induced by an ideal of B .*
- (iii) *i is injective and $\text{Coker } i$ is a flat A -module.*
- (iv) *The functor $M_{(B)} = M \otimes_A B$ in the A -module M is exact, and the canonical functorial homomorphism $M \rightarrow M_{(B)}$ is injective.*
- (iv bis) *For every ideal I of A , $I \otimes_A B \rightarrow IB$ is an isomorphism, and the inverse image of IB in A is equal to I .*

One has (i) $\Rightarrow$ (ii) by 2.3, (ii) $\Rightarrow$ (ii bis) is trivial, (ii bis) $\Rightarrow$ (i) by criterion (i ter) of 2.1. One has (iii) $\Rightarrow$ (iv) by 1.1, (iv) $\Rightarrow$ (iv bis) trivially (by taking $M = A/I$ in the second condition (iv bis)), and (iv bis) $\Rightarrow$ (i) by virtue of the criterion of flatness by ideals seen at the beginning of 1 and of criterion 2.1 (i ter). Finally, (iv) $\Rightarrow$ (iii) by an easy converse of 1.1, and (i) $\Rightarrow$ (iv) since if N is the kernel of $M \rightarrow M \otimes_A B = T(M)$, then (T being exact) $N \rightarrow T(N)$ is zero, whence $T(N) = N \otimes_A B = 0$, whence $N = 0$, qed.

3. Relations with completion

Let A be a noetherian ring, I an ideal in A , $\hat{A}$ the separated completion of A for the I -preadic topology, and for every A -module M , let $\hat{M}$ be its completion for the I -preadic topology. It is an $\hat{A}$ -module, whence a canonical homomorphism

$$M \otimes_A \hat{A} \longrightarrow \hat{M}.$$

When M runs through the modules of finite type, the functor $M \mapsto \hat{M}$ is exact, as follows easily from Krull's theorem: If $N \subset M$, the topology of N is the one induced by the topology of M . Since $M \otimes_A \hat{A}$ is right exact, one easily concludes from this (by resolving M by $L \rightarrow L' \rightarrow M$, with L and L' free of finite type) that the functorial homomorphism above is an isomorphism ($\hat{M}$ being also right exact) and consequently that $M \otimes_A \hat{A}$ is also an exact functor in M . Hence:

PROPOSITION 3.1. *Let A be a noetherian ring, I an ideal of A ; then the separated completion $\hat{A}$ of A (for the I -preadic topology) is flat over A .*

COROLLARY 3.2. *For $\hat{A}$ to be faithfully flat over A , it is necessary and sufficient that I be contained in the radical of A .*

Indeed, it suffices to apply criterion 2.1 (i ter).

These results summarize all that one knows how to say, from the point of view of linear algebra, about the relations between A and $\hat{A}$. Corollary 3.2 is used above all when A is a noetherian local ring and I is contained in the maximal ideal $\mathfrak{m}$ (and most often, is equal to it).

4. Relations with free modules

PROPOSITION 4.1. *Let A be a ring, I an ideal of A , M an A -module. Suppose that one is under one or the other of the following hypotheses:*

- (a) *I is nilpotent*
- (b) *A is noetherian, I is in the radical of A , and M is of finite type.*

For M to be free over A , it is necessary and sufficient that $M \otimes A/I$ be free over A/I , and that $\text{Tor}_1^A(M, A/I) = 0$.

It is necessary; let us prove sufficiency. Let (e_i) be a family of elements of M whose image in $M \otimes A/I = M/IM$ defines a basis of it over A/I (it is a finite family in case (b)). Let L be the free A -module constructed on the same index set; one thus has a homomorphism $L \rightarrow M$ such that tensoring T with A/I induces an isomorphism $T(L) \xrightarrow{\sim} T(M)$. If Q is the cokernel of $L \rightarrow M$, one therefore has $T(Q) = 0$, whence $Q = 0$ by virtue of Nakayama (valid under one or the other condition (a) or (b)). Hence $L \rightarrow M$ is surjective; let R be its kernel; one thus has an exact sequence

$$0 \longrightarrow R \longrightarrow L \longrightarrow M \longrightarrow 0$$

whence, since $\text{Tor}_1^A(M, A/I) = 0$, an exact sequence $0 \rightarrow T(R) \rightarrow T(L) \rightarrow T(M) \rightarrow 0$, whence $T(R) = 0$, whence further $R = 0$ by virtue of Nakayama (taking into account that in case (b), R is of finite type since A was assumed noetherian).

COROLLARY 4.2. *One can replace the condition $\text{Tor}_1^A(M, A/I) = 0$ by: the canonical surjective homomorphism*

$$(*) \quad \text{gr}_I^0(M) \otimes_{A/I} \text{gr}_I(A) \longrightarrow \text{gr}_I(M)$$

is an isomorphism.

Indeed, if M is free, this is certainly satisfied. One must therefore prove that if $M \otimes A/I$ is free over A/I and the condition on the gr is satisfied, then M is free. One takes up the proof above again, constructing $L \rightarrow M$; it follows from the hypothesis that this homomorphism induces an isomorphism for the associated graded modules, hence its kernel is contained in the intersection of the $I^n L$, hence is zero (as is trivial in (a), and well known in (b)). Qed.

COROLLARY 4.3. *Suppose that A/I is a field. Then the following conditions on M are equivalent:*

- (i) *M is free*
- (ii) *M is projective*
- (iii) *M is flat*
- (iv) *$\text{Tor}_1^A(M, A/I) = 0$*
- (v) *The canonical homomorphism $(*)$ is bijective.*

Indeed, in the case considered, $M \otimes A/I$ is automatically free.

The preceding result is valid in the following two cases:

- (a) *M is an arbitrary module over a local ring A whose maximal ideal I is nilpotent (for example an artinian local ring).*
- (b) *M is a module of finite type over a noetherian local ring.*

Let us recall, for the record:

COROLLARY 4.4. *Suppose that A is a noetherian local integral ring with maximal ideal $\mathfrak{m} = I$, residue field $k = A/I$, field of fractions K . Let M be a module of finite type over A . Then the equivalent conditions (i) to (v) above are also equivalent to*

(vi) *$M \otimes_A K$ and $M \otimes_A k$ are vector spaces of the same dimension (i.e. the rank of M over A is equal to the minimum number of generators of the A -module M).*

Immediate proof; we leave to the reader the task of generalizing to the case where A is only assumed to be without nilpotent elements: one must then require that the ranks of M at the minimal prime ideals of A be equal to the dimension of the vector space $M \otimes_A k$.

5. Local criteria of flatness

PROPOSITION 5.1. *Let A be a ring equipped with an ideal I , M an A -module. Suppose*

$$\mathrm{Tor}_1^A(M, A/I^n) = 0 \quad \text{for } n > 0$$

then the canonical surjective homomorphism

$$(*) \quad \mathrm{gr}_I^0(M) \otimes_{A/I} \mathrm{gr}_I(A) \longrightarrow \mathrm{gr}_I(M)$$

is an isomorphism. The converse is true if I is nilpotent.

The hypothesis means that the homomorphisms

$$I^n \otimes_A M \longrightarrow I^n M$$

are isomorphisms, whence at once the fact that the homomorphisms

$$I^n/I^{n+1} \otimes_A M \longrightarrow I^n M/I^{n+1} M$$

are isomorphisms. Conversely, suppose this condition satisfied and I nilpotent; let us prove $\mathrm{Tor}_1^A(M, A/I^n) = 0$ for every n . It is true for n large; let us proceed by descending induction on n , supposing it proved for $n+1$. We have a commutative diagram

$$\begin{array}{ccccccc} M \otimes I^{n+1} & \longrightarrow & M \otimes I^n & \longrightarrow & M \otimes (I^n/I^{n+1}) & \longrightarrow & 0 \\ \downarrow & & \downarrow & & \downarrow & & \\ 0 & \longrightarrow & MI^{n+1} & \longrightarrow & MI^n & \longrightarrow & MI^n/MI^{n+1} \longrightarrow 0 \end{array}$$

where the rows are exact. By hypothesis, the last vertical arrow is an isomorphism, and the induction hypothesis also means that the first vertical arrow is one. Hence the same holds for the middle vertical arrow, which completes the proof.

The following proposition was brought out at the time of the Seminar by Serre; it allows substantial simplifications in the present no.

PROPOSITION 5.2. *Let $A \rightarrow B$ be a ring homomorphism, M an A -module. The following conditions are equivalent:*

- (i) *For every B -module N , one has $\mathrm{Tor}_1^A(M, N) = 0$;*
- (ii) *$\mathrm{Tor}_1^A(M, B) = 0$, and $M_{(B)} = M \otimes_A B$ is B -flat.*

We have a functorial isomorphism

$$M \otimes_A N = (M \otimes_A B) \otimes_B N$$

which expresses the left-hand side, considered as a functor in M , as a composite of two functors $M \mapsto M \otimes_A B$ and $P \mapsto P \otimes_B N$. As the first transforms free modules over A into free modules over B , hence projectives into projectives, we have the spectral sequence of composite functors

$$\mathrm{Tor}_n^A(M, N) \longleftarrow \mathrm{Tor}_p^B(\mathrm{Tor}_q^A(M, B), N)$$

whence an exact sequence for the terms of low degree

$$0 \longleftarrow \mathrm{Tor}_1^B(M \otimes_A B, N) \longleftarrow \mathrm{Tor}_1^A(M, N) \longleftarrow \mathrm{Tor}_1^A(M, B) \otimes_A N$$

If (i) is satisfied, then one concludes from this exact sequence $\mathrm{Tor}_1^B(M \otimes_A B, N) = 0$ for every N , i.e. $M \otimes_A B$ is B -flat, whence (ii). If conversely (ii) is satisfied, then in the exact sequence the terms surrounding $\mathrm{Tor}_1^A(M, N)$ are zero, hence we have (i).

COROLLARY 5.3. *Suppose that $B = A/I$; then the preceding conditions are equivalent to the following one:*

(iii) $\mathrm{Tor}_1^A(M, N) = 0$ for every A -module N annihilated by a power of I .

Indeed, (i) means that this is so if N is annihilated by I . One deduces (iii) from this by applying the hypothesis to the $I^n N / I^{n+1} N$.

COROLLARY 5.4. *Under the conditions of 5.3, the conditions considered imply that the functorial homomorphism*

$$(*) \quad \mathrm{gr}_I^0(M) \otimes_{A/I} \mathrm{gr}_I(A) \longrightarrow \mathrm{gr}_I(M)$$

is an isomorphism, and that $M \otimes_A A/I$ is flat over A/I .

It suffices to apply (iii) and (5.1). Using the converse of 5.1 in the case I nilpotent, one finds:

COROLLARY 5.5. *Let A be a ring equipped with a nilpotent ideal I , M an A -module. The following conditions are equivalent:*

- (i) M is A -flat
- (ii) $M \otimes_A A/I$ is A/I -flat, and $\mathrm{Tor}_1^A(M, A/I) = 0$
- (iii) $M \otimes_A A/I$ is A/I -flat, and the canonical homomorphism $(*)$ on the graded modules is an isomorphism.

Indeed, these are respectively the preceding conditions (iii) and (ii), and those of corollary 5.4.

Let us no longer suppose I nilpotent; then we shall only have a priori in 5.5 the implications (i) $\Rightarrow$ (ii) $\Rightarrow$ (iii). On the other hand, as condition (iii) remains stable on dividing by a power of I , we see by virtue of 5.5 that it implies

(iv) for every integer n , $M \otimes A/I^n$ is flat over A/I^n .

We propose to give conditions under which one can conclude from this (i), i.e. that M is A -flat. I say that it suffices for this that A be noetherian and that M satisfy the following finiteness condition: *for every module of finite type N over A , $M \otimes_A N$ is separated for the I -preadic topology.* (It would suffice to verify it if N is an ideal of finite type in A). Indeed, let us prove that under these conditions, if $N' \rightarrow N$ is a monomorphism of modules of finite type, $M \otimes_A N' \rightarrow M \otimes_A N$ is a monomorphism. It suffices indeed to show that the kernel is contained in the $I^n(M \otimes_A N') = \mathrm{Im}(M \otimes_A I^n N' \rightarrow M \otimes_A N')$, or again in the $\mathrm{Im}(M \otimes_A V'_n \rightarrow M \otimes_A N') = \mathrm{Ker}(M \otimes_A N' \rightarrow M \otimes_A (N'/V'_n))$, where V'_n runs through a countable fundamental system of neighborhoods of 0 in N' (equipped with its I -adic topology). By Krull's theorem, the I -adic topology of N' is induced by that of N , so we can take $V'_n = N' \cap I^n N$. Consider then the commutative diagram

$$\begin{array}{ccc} M \otimes_A N' & \longrightarrow & M \otimes_A (N'/V'_n) \\ \downarrow & & \downarrow \\ M \otimes_A N & \longrightarrow & M \otimes_A (N/I^n N) \end{array}$$

As N'/V'_n and $N/I^n N$ are annihilated by I^n , the second vertical homomorphism is identified with the one deduced from the *injective* homomorphism $N'/V'_n \rightarrow N/I^n N$ by tensoring over A/I^n with the *flat* (A/I^n) -module $M \otimes_A A/I^n$; it is therefore *injective*. Consequently, the kernel of $M \otimes_A N' \rightarrow M \otimes_A N$ is contained in that of $M \otimes_A N' \rightarrow M \otimes_A (N'/V'_n)$, which is what we wanted.

The “finiteness” condition considered on M is satisfied in particular if M is a module of finite type over a noetherian A -algebra B such that IB is contained in the radical of B : indeed, then $M \otimes_A N$ is a module of finite type over B for every module of finite type N over A , hence separated by Krull for the I -adic topology = its (IB) -adic topology. We thus find:

THEOREM 5.6. *Let $A \rightarrow B$ be a homomorphism of noetherian rings, I an ideal of A such that IB is contained in the radical of B , M a B -module of finite type. The following conditions are equivalent:*

- (i) M is A -flat
- (ii) $M \otimes_A A/I$ is A/I -flat, and $\mathrm{Tor}_1^A(M, A/I) = 0$
- (iii) $M \otimes_A A/I$ is A/I -flat, and the canonical homomorphism

$$\mathrm{gr}_I^0(M) \otimes_{A/I} \mathrm{gr}_I(A) \longrightarrow \mathrm{gr}_I(M)$$

is an isomorphism.

- (iv) For every integer n , $M \otimes_A A/I^n$ is flat over A/I^n .

This result applies above all when A, B are noetherian *local* rings, $A \rightarrow B$ a local homomorphism, and I an ideal of A contained in its maximal ideal (and one can immediately reduce 5.6 to this case). An interesting case is that where A/I is a field, i.e. I maximal, in which case the condition that $M \otimes_A (A/I)$ be flat over A/I becomes unnecessary; moreover, as then the A/I^n are artinian local rings, condition (iv) means that the $M \otimes_A (A/I^n)$ are free over the A/I^n .

COROLLARY 5.7. *Let $A \rightarrow B$ be a local homomorphism of noetherian local rings, $u: M' \rightarrow M$ a homomorphism of B -modules of finite type; suppose M flat over A . Then the following conditions are equivalent:*

- (i) u is injective, and $\mathrm{Coker} u$ is flat over A .
- (ii) $u \otimes_A k: M' \otimes_A k \rightarrow M \otimes_A k$ is injective

(where k denotes the residue field of A).

(i) $\Rightarrow$ (ii) by virtue of 1.1; let us prove the converse. First of all u is injective, for it suffices to verify this on the associated graded modules, where it follows from a commutative square that the reader will write down. Let M'' be its Coker; we thus have an exact sequence

$$0 \longrightarrow M' \longrightarrow M \longrightarrow M'' \longrightarrow 0$$

whence by the exact sequence of the Tor, taking into account hypothesis (ii) and $\mathrm{Tor}_1^A(M, k) = 0$, the relation $\mathrm{Tor}_1^A(M'', k) = 0$, hence M'' is flat over A by theorem 5.6.

COROLLARY 5.8. *Under the conditions of 5.6, let J be an ideal of B containing IB and contained in the radical. Let $\hat{A}$ be the I -adic completion of A and $\hat{B}$ and $\hat{M}$ the J -adic completions of B and M . For M to be A -flat, it is necessary and sufficient that M be $\hat{A}$ -flat.*

(N.B. the sufficiency would already follow easily from 3.2). We use criterion (iii) of 5.6 in the situation (A, B, I, M) and in the situation $(\hat{A}, \hat{B}, I\hat{A}, \hat{M})$. One observes that the conditions obtained for the one and the other case are equivalent, thanks to 3.2.

COROLLARY 5.9. *Let $A \rightarrow B \rightarrow C$ be local homomorphisms of noetherian local rings, M a C -module of finite type (N.B. C intervenes only in order to be able to put a finiteness condition on M). We suppose B flat over A . Let k be the residue field of A . Equivalent conditions:*

- (i) M is flat over B .
- (ii) M is flat over A , and $M \otimes_A k$ is flat over $B \otimes_A k$.

The implication (i) $\Rightarrow$ (ii) is trivial; let us prove (ii) $\Rightarrow$ (i). We apply criterion (iii) of 5.6 to $(B, C, \mathfrak{m}_B(B/I) = M \otimes_B (B \otimes_A k) = M \otimes_A k$, the first condition of this

criterion means precisely that $M \otimes_A k$ is flat over $B \otimes_A k$, fine. The second condition of the criterion is satisfied because M is flat over A and B flat over A , by an associativity formula for the tensor product. — Of course, referring to 5.5 instead of 5.6, one obtains an analogous statement without noetherian and finiteness conditions, when one supposes on the other hand that the ideal $\mathfrak{m}$ of A is nilpotent. (The fact that $\mathfrak{m}$ was taken maximal has moreover not intervened; but it is in a sense the case “ $\mathfrak{m}$ maximal” which is “the best possible”).

6. Flat morphisms and open sets

Let us first recall some results on constructible sets, which are moreover proved in notes in circulation of the Dieudonné-Rosenlicht Seminar on Schemes⁴.

Let X be a topological space. We say with Chevalley that a subset of X is *constructible* if it is a finite union of locally closed subsets.

LEMMA 6.1. *Let X be a noetherian topological space, let Z be a subset of X . For Z to be constructible, it is necessary and sufficient that for every irreducible closed subset Y of X , $Z \cap Y$ be non-dense in Y or contain a nonempty open subset of the space Y .*

One deduces from it, using a well-known lemma of Commutative Algebra:

LEMMA 6.2 (Chevalley). *Let $f: X \rightarrow Y$ be a morphism of finite type of preschemes, with Y noetherian. Then $f(X)$ is constructible.*

LEMMA 6.3. *Let X be a noetherian topological space every irreducible closed subset of which admits a generic point, U a constructible subset of X , $x \in X$. For U to be a neighborhood of x , it is necessary and sufficient that every generization y of x (i.e. every $y \in X$ such that $x \in \overline{y}$) be in U .*

In particular

COROLLARY 6.4. *Let X be a noetherian topological space every irreducible closed subset of which admits a generic point, U a subset of X . For U to be open, it is necessary and sufficient that it satisfy the following two conditions:*

- (a) *U contains every generization of each of its points*
- (b) *if $x \in U$, then $U \cap \overline{x}$ contains a nonempty open subset of the space $\overline{x}$.*

Indeed, U is necessarily constructible thanks to 6.1, and we apply criterion 6.2, which proves that U is a neighborhood of each of its points.

COROLLARY 6.5. *Let $f: X \rightarrow Y$ be a morphism of finite type of preschemes, with Y locally noetherian, x a point of X , $y = f(x)$. For f to transform every neighborhood of x into a neighborhood of y , it is necessary and sufficient that for every generization y' of y , there exist a generization x' of x such that $f(x') = y'$.*

We can evidently suppose that X and Y are affine, hence noetherian. The condition is sufficient, for it suffices to prove that $f(X)$ is a neighborhood of y ; now $f(X)$ is constructible by 6.1, and it suffices to apply criterion 6.3. The condition is necessary, for let $Y' = \overline{y'}$, and let F be the union of the irreducible components of $f^{-1}(Y')$ which do not contain x . Then $X - F$ is an open neighborhood of x , hence its image is a neighborhood of y , and a fortiori contains y' , hence there exists $x'_1 \in X - F$ such that $f(x'_1) = y'$. Consider an irreducible component of $f^{-1}(Y')$ containing x'_1 ; it necessarily contains x (for otherwise it would be contained in F); let x' be its generic point. It is a generization of x , and $f(x')$ is a generization of $f(x'_1) = y'$ contained in Y' , hence is equal to y' , qed.

THEOREM 6.6. *Let $f: X \rightarrow Y$ be a morphism locally of finite type, with Y locally noetherian, F a coherent sheaf on X with support X , flat with respect to Y . Then f is an open morphism (i.e., transforms open sets into open sets).*

⁴Cf. EGA 0_{III} 9, EGA IV 1.8 and 1.10.

It suffices to prove criterion 6.5 for every point $x \in X$. Now the generizations x' of x correspond to the prime ideals of $\mathcal{O}_x$, those y' of y correspond to the prime ideals of $\mathcal{O}_y$, and one must therefore verify that every prime ideal of $\mathcal{O}_y$ is induced by a prime ideal of $\mathcal{O}_x$. Now F_x is a nonzero $\mathcal{O}_x$ -module and $\mathcal{O}_y$ -flat, hence faithfully flat over $\mathcal{O}_y$ by 2.2. We can therefore apply 2.3, which completes the proof.

REMARKS. As flatness is preserved by extension of the base, we see that under the conditions of 6.5 f is even *universally open*. I do not know however, when Y is integral and X of finite type over Y , whether f induces on every component X_i of X an open morphism, or even only an equidimensional one⁵, i.e. one all of whose fiber components have the same dimension (one knows only that X_i *dominates* Y). The question is linked to the following: let $A \rightarrow B$ be a local homomorphism of noetherian local rings, such that B is flat over A and $\mathfrak{m}B$ is an ideal of definition of B (which moreover implies $\dim B = \dim A$). Is it true that for every minimal prime ideal $\mathfrak{p}_i$ of B , one has $\dim B/\mathfrak{p}_i = \dim B$? Let us point out only that the answer to the first question is negative when one replaces the flatness hypothesis of 6.5 by the sole hypothesis that f be universally open.

LEMMA 6.7. *Let A be a noetherian integral domain, B an A -algebra of finite type, M a B -module of finite type. Then there exists a nonzero element f of A such that M_f is a free (a fortiori flat) module over A_f .*

Let K be the field of fractions of A ; then $B \otimes_A K$ is an algebra of finite type over K , and $M \otimes_A K$ a module of finite type over the latter. Let n be the dimension of the support of this module; we shall reason by induction on n . If $n < 0$, i.e. if $M \otimes_A K = 0$, then taking a finite number of generators of M over B , we see that there exists an $f \in A$ which annihilates these generators, hence M , whence $M_f = 0$ and we have won. Suppose $n \geq 0$. We know that the B -module M admits a composition series whose successive quotients are isomorphic to modules $B/\mathfrak{p}_i$, the $\mathfrak{p}_i$ being prime ideals of B . As an extension of free modules is free, we are reduced to the case where M itself is of the form $B/\mathfrak{p}$, or again identical to B , B being an A -algebra that is an *integral domain*. Applying the Noether normalization lemma to the K -algebra $B \otimes_A K$, we see easily that there exists a nonzero element f of A such that B_f is integral over the subring $A_f[t_1, \dots, t_n]$, where the t_i are indeterminates. Hence we can already suppose B is integral over $C = A[t_1, \dots, t_n]$; it is therefore a torsion-free C -module of finite type. Let m be its rank; there exists therefore an exact sequence of C -modules:

$$0 \longrightarrow C^m \longrightarrow B \longrightarrow M' \longrightarrow 0$$

where M' is a torsion C -module. It follows that the Krull dimension of the $C \otimes_A K$ -module $M' \otimes_A K$ is strictly less than the dimension n of $C \otimes_A K$. By the induction hypothesis, it follows that, on condition of localizing with respect to a suitable nonzero f of A , we can suppose that M' is a free A -module. On the other hand, C^m is a free A -module. Hence B is then a free A -module; we are done.

LEMMA 6.8. *Let A be a noetherian ring, B an algebra of finite type over A , M a B -module of finite type, $\mathfrak{p}$ a prime ideal of B , $\mathfrak{q}$ the prime ideal that it induces on A . We suppose $M_{\mathfrak{p}}$ flat over $A_{\mathfrak{q}}$ (or over A , it is the same thing). Then there exists a $g \in B - \mathfrak{p}$ such that*

- (a) $(M/\mathfrak{q}M)_g$ is flat over $A/\mathfrak{q}$.
- (b) $\mathrm{Tor}_1^A(M, A/\mathfrak{q})_g = 0$.

Indeed, applying 6.7 to $(A/\mathfrak{q}, B/\mathfrak{q}B, M/\mathfrak{q}M)$ we see first that there exists an f in $A - \mathfrak{q}$ such that $(M/\mathfrak{q}M)_f$ is flat over $A/\mathfrak{q}$. On the other hand, as $M_{\mathfrak{p}}$ is flat over A , we have $\mathrm{Tor}_1^A(M, A/\mathfrak{q})_{\mathfrak{p}} = \mathrm{Tor}_1^A(M_{\mathfrak{p}}, A/\mathfrak{q}) = 0$, hence, as $\mathrm{Tor}_1^A(M, A/\mathfrak{q})$ is a B -module of finite type, there exists a $g \in B - \mathfrak{p}$ such that we have (b). We can then (replacing g by gf) suppose that we have at the same time (a), which proves the corollary.

⁵The answer to the second question is affirmative, the answer to the first negative even if f is étale; cf. EGA IV 12.1.1.5 and EGA ErrIV 33.

COROLLARY 6.9. *With the notation of 6.8, for every prime ideal $\mathfrak{p}'$ of B containing $\mathfrak{p}$ and not containing g , $M_{\mathfrak{p}'}$ is flat over A (or, what amounts to the same, over $A_{\mathfrak{q}'}$, where $\mathfrak{q}'$ is the prime ideal of A induced by $\mathfrak{p}'$).*

It suffices to apply criterion 5.6 (ii) to the system $(A, B_{\mathfrak{q}'}, \mathfrak{q}, M_{\mathfrak{q}'})$, using the localization of the Tor.

THEOREM 6.10. *Let $f: X \rightarrow Y$ be a morphism of finite type, with Y locally noetherian, and let F be a coherent sheaf on X . Let U be the set of points $x \in X$ such that F_x is flat over $\mathcal{O}_{f(x)}$. Then U is an open set.*

Proof. We can suppose X and Y affine, with rings B and A , hence F defined by a B -module M of finite type. We apply criterion 6.4. Condition (a) is verified trivially by 1.2 (i); it remains to verify condition (b) of 6.4. This is what was done in lemma 6.8 and corollary 6.9.

In many questions, the following weaker form of theorem 6.10 is sufficient (which already follows from lemma 6.7, and therefore requires neither the technique of constructibles, nor theorem 5.6):

COROLLARY 6.11. *Under the conditions of 6.10, if we suppose Y integral, then there exists a nonempty open set V in Y such that F is flat relative to Y at all points of $f^{-1}(V)$.*

Indeed, the open set U contains the fiber of the generic point of Y (since the local ring of this point is a field), hence it contains an open set of the form $f^{-1}(V)$, X being of finite type over Y . From 6.11, one also easily concludes the following result, where Y is assumed noetherian (but not necessarily integral): there exists a partition of Y into locally closed subsets Y_i such that (equipping Y_i with the induced reduced structure) F induces on each $X_i = X \times_Y Y_i$ a sheaf flat with respect to Y_i .